

VASAVI COLLEGE OF ENGINEERING

(AUTONOMOUS)
(Affiliated to Osmania University)

Hyderabad - 500 031.

DEPARTMENT OF

IT

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AIML

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Week 1: Numpy, scipy and Matplotlib:

NUMPY

- it's a library in python which contains several useful numerical functions which are useful and not available in standard library.
1. `import numpy as np`
 2. `zero = np.zeros((1,20))` # create an array of 20 zeroes
`print(zero)`
 3. `one = np.ones((1,20))` # create an array of 20 ones
`print(one)`
 4. `np.full((1,10), 5)` # create an array of 10 fives
 5. `np.arange(5,100)` # create an array of the integers from 5 to 100.
 6. `np.arange(10,50,2)` # create an array of all the even integers 10 to 50
 7. `a = np.arange(0,25)` # create a 5x5 matrix ranging [0,24]
`a.resize((5,5))`
`print(a)`
 8. `np.eye(5,5)` # create a 5x5 identity matrix.
 9. `np.random.randint(0,2)` # generate a random number btw 0 & 1
 10. `np.linspace(0,1,5)` # create an array of 5 linearly spaced points btw 0,1.
 11. `matrix = np.arange(1,26).reshape(5,5)`
`matrix`
 12. `matrix[2:,1:]` # indexing
 13. `matrix[3,4]` # selection.

NUMPY
outputs:

2. `[0. 0. 0. 0. 0. 0. 0. 0. 0. 0. 0. 0. 0. 0. 0. 0. 0. 0. 0. 0.]`
3. `[1. 1. 1. 1. 1. 1. 1. 1. 1. 1. 1. 1. 1. 1. 1. 1. 1. 1. 1. 1.]`
4. `array([5, 5, 5, 5, 5, 5, 5, 5, 5, 5])`
5. `array([5, 6, 7, 8, 9, 10, ..., 95, 96, 97, 98, 99, 100])`
6. `array([10, 12, 14, 16, 18, 20, 22, 24, 26, ..., 44, 46, 48])`
7. `[0 1 2 3 4]`
`[5 6 7 8 9]`
`[10 11 12 13 14]`
`[15 16 17 18 19]`
`[20 21 22 23 24]`
8. `array([[1., 0., 0., 0., 0.],
[0., 1., 0., 0., 0.],
[0., 0., 1., 0., 0.],
[0., 0., 0., 1., 0.],
[0., 0., 0., 0., 1.]])`
9. 0
10. `array([0., 0.25, 0.5, 0.75, 1.])`
11. `array([1, 2, 3, 4, 5],
[6, 7, 8, 9, 10],
[11, 12, 13, 14, 15],
[16, 17, 18, 19, 20],
[21, 22, 23, 24, 25]])`
12. `array([[12, 13, 14, 15],
[17, 18, 19, 20],
[22, 23, 24, 25]])`
12 array
13. 20

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14. `a = matrix([0:3, 1])`

`a.reshape(3, 1)`

15. `matrix([4, :])`

16. `matrix([3, :])`

NUMPY OPERATIONS:

17. `np.sum(matrix)` # get sum of all the values in matrix

18. `matrix.std()` # get standard deviation of the values in matrix

19. `np.sum(matrix, axis = 0)` # get sum of all the columns in matrix

20. `A = np.array([[2.0, 4.0, 7.0], [3.0, 3.0, 2.0], [5.0, 6.0, 3.0]])`

`B = np.array([4.0, 8.0, 0.0])`

`np.linalg.solve(A, B)` # solve the above linear algebra.

MATPLOTLIB

→ This python library is useful when a user wants to graphically represent the data in the form of plots, graphs,

1. `import matplotlib.pyplot as plt`

`x = np.linspace(0, 2, 100)`

`plt.plot(x, x, label = 'linear')`

`plt.plot(x, x**2, label = 'quadratic')`

`plt.plot(x, x**3, label = 'cubic')`

`plt.xlabel('x label')`

`plt.ylabel('y label')`

`plt.title('Simple plot')`

`plt.legend()`

`plt.show()`

14. array ([[2],
[7],
[12]])

15. array([21, 22, 23, 24, 25])

16. array([[16, 17, 18, 19, 20],
[21, 22, 23, 24, 25]])

17. 325

18. 7.211102550927978

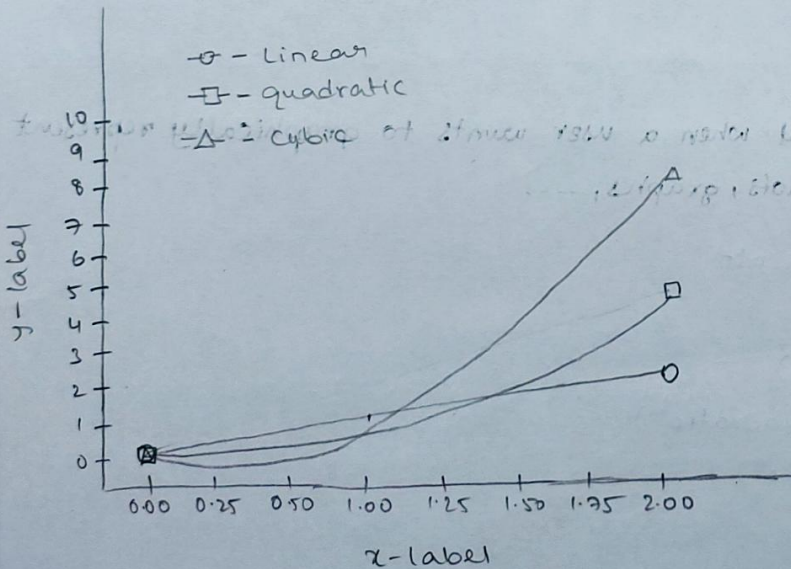
19. array([55, 60, 65, 70, 75])

20. array([12., -12., 4.]).

MATPLOTLIB

outputs:

Simple Plot



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SCIPY

→ ~~It~~ it stands for scientific python and has many important scientific functions which are not ~~not~~ available in numpy.

1. From scipy import linalg

```
z = np.array([[2.0, 4.0, 7.0], [3.0, 3.0, 2.0], [5.0, 6.0, 3.0]])
```

```
eg_val, eg_vect = linalg.eig(z)
```

```
print("Eigen values: \n", eg_val)
```

```
print("Eigen vector: \n", eg_vect)
```

2.

```
a = np.array([1.0, 2.0], [1.0, 4.0])
```

```
print(linalg.inv(a))
```

```
print(linalg.det(a))
```

```
print(a.dot(linalg.inv(a)))
```

SCIPY

outputs:

$$2. a. \begin{bmatrix} -2. & 1. \\ 1.5 & -0.5 \end{bmatrix}$$

$$b. -20$$

$$x. \begin{bmatrix} 1.0000000e+00 & 0.0000000e+00 \\ 8.8817842e-16 & 1.0000000e+00 \end{bmatrix}$$

1. Eigen values:

$$[11.46042088+0.j \quad -2.88595626+0.j \quad -0.57446462+0.j]$$

Eigen vector:

$$[-0.04859937 \quad -0.85462715 \quad 0.72657746]$$

$$[-0.38517552 \quad 0.28901623 \quad -0.67662652]$$

$$[-0.85648443 \quad 0.43137228 \quad 0.11942259]$$

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VIVA QUESTIONS

1. Explain features of Numpy.

A. Numpy provides efficient, multi-dimensional arrays, which support fast mathematical operations and array broadcasting. Arrays are stored in contiguous memory, making computations much faster than with python lists.

2. Explain features of Scipy Library.

A. Scipy is built atop numpy, offering advanced, scientific computing models for optimization, integrations, linear algebra, image processing and statistics.

3. Explain features of Matplotlib

A. Matplotlib enables the creation of diverse, high-quality static and interaction visualizations such as line plots, bar charts, histograms and 3D plots. Plots are cross-platform compatible and can be publication-ready.

4. Explain features of Scikit library.

A. Scikit-learn is an open-source machine-learning library that provides a vast suite of tools for classification, regression, clustering and preprocessing. Features include supervised and unsupervised learning algorithms, dimensionality reduction (eg. PCA), model selection and evaluation metrics, all via a consistent and easy-to-use API.

5. Numpy NDArray Usages.

A. Vectorized math, matrix operations, statistical analysis, reshaping, slicing, advanced indexing and broadcasting for element-wise operations. Attributes include (.shape), (.ndim), (.size), (.dtype), and (.itemsize) for efficient manipulation and querying.

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6. Overview of Matplotlib API interface.

A. The API supports stepwise, flexible plot building and fine customization at every stage.